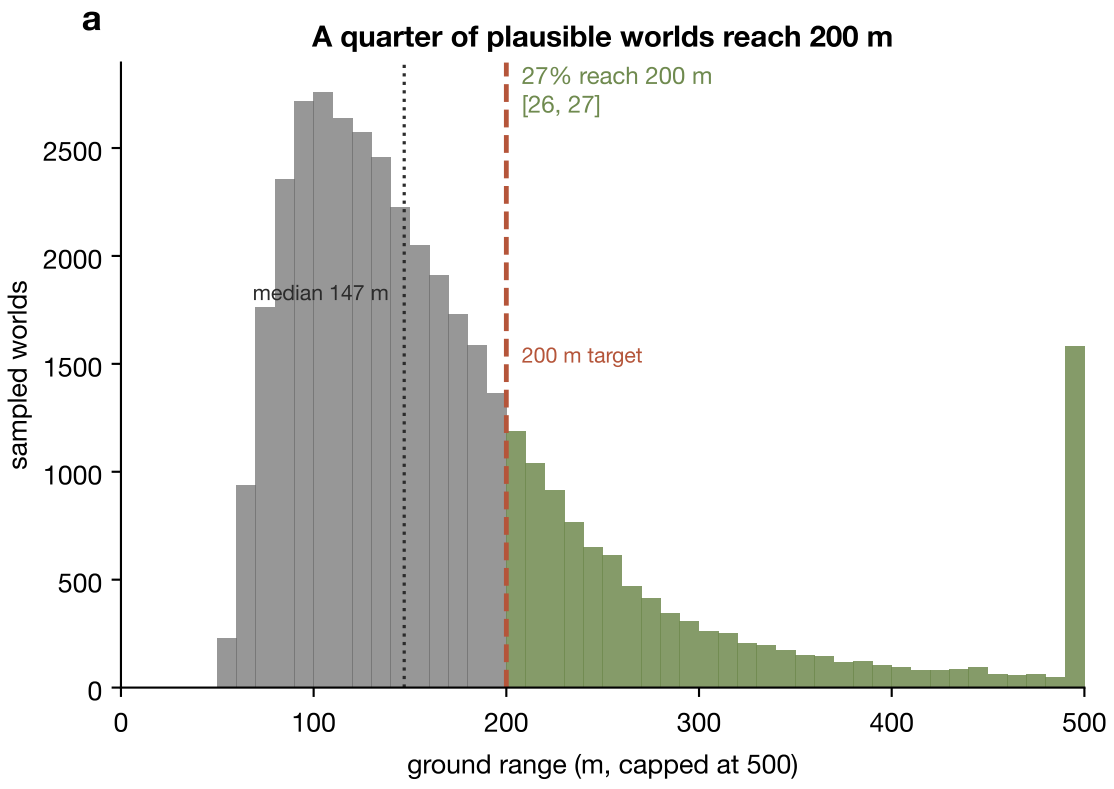
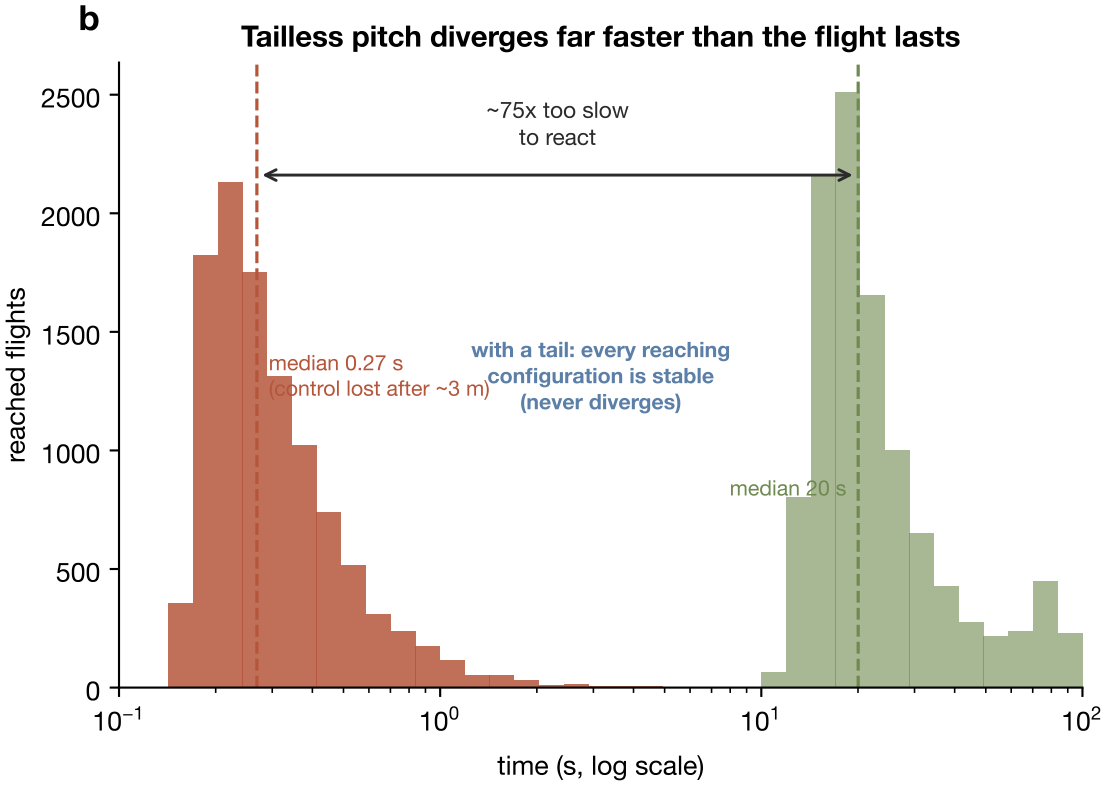

a
A quarter of plausible worlds reach 200 m
27% reach 200 m
[26, 27]
median 147 m
200 m target
sampled worlds
ground range (m, capped at 500)
0
500
1000
1500
2000
2500
0
100
200
300
400
500


b
Tailless pitch diverges far faster than the flight lasts
~75x too slow
to react
median 0.27 s
(control lost after ~3 m)
with a tail: every reaching
configuration is stable
(never diverges)
median 20 s
reached flights
time (s, log scale)
0
500
1000
1500
2000
2500
10⁻¹
10⁰
10¹
10²
no tail: time to lose pitch control
flight duration (time it must stay controlled)